



Guidelines

Vein Selection and Preservation in Patients with Chronic Kidney Disease

Reference #: CPG019

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	All Staff

Introduction

Patients with stage 3b-4 chronic kidney disease (CKD) are at increased risk of progression to end stage kidney disease requiring renal replacement therapy (RRT). The ability to establish permanent vascular access for haemodialysis is largely dependent on the presence of suitable peripheral veins. The preservation of veins to allow for permanent vascular access can be assisted by reducing injury to vessels caused by frequent venepuncture and indiscriminate use of intravascular devices into central or peripheral veins.

In life threatening situations where urgent vascular access is required the easiest available option should be used.

Stage	GFR (ml/min/1.73m ²)	Description
G1	>90	Normal or high
G2	60-89	Mildly decreased
G3a	45-59	Mildly to moderately decreased
G3b	30-44	Moderately to severely decreased
G4	15-29	Severely decreased
G5	<15	Kidney failure

Table 1: Stages of Chronic Kidney Disease based on updated Kidney Disease Improving Global Outcomes (KDIGO) ¹

This guideline is based on The WA Renal Health Network's 2017 [Guideline for Vein Preservation in Chronic Kidney Disease](#),² which recommends vein selection practices for patients with CKD.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☐	Staff Safety	☐
Other:	☐		

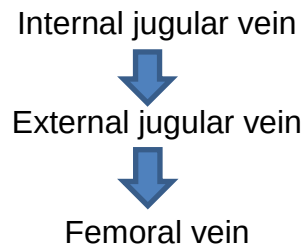
1. Vein Selection and Preservation

1.1 Peripheral Vascular Access

- Select dorsal **veins of the dominant hand** when possible.
- If not possible refer to the vein selection hierarchy in [Appendix 1](#)
- Ensure tourniquet is not applied on or above a functioning vascular access.

1.2 Central Venous Access

- **Avoid** placement of non-tunnelled central venous catheter (CVC), particularly in the subclavian vein in **all patients with CKD** if possible
- If unavoidable the preferred site for central venous access in **decreasing order** is:



1.3 Prolonged Intravenous Access

Avoid peripherally inserted catheter (PICC) and midlines in **all patients with CKD**. Only consider if there are no other options.

A small-bore CVC in the internal jugular vessels is preferable if prolonged intravenous access is required.

1.4 Antibiotic Use

Appropriate oral antibiotic alternatives should be considered for all patients with CKD, whenever possible and medically suitable. In haemodialysis patients, consider the use of longer acting, renally cleared intravenous antibiotics particularly those, which can be delivered during dialysis. Dose adjustment may be recommended in consultation with renal pharmacist. Consultation with the infectious disease team is recommended.

1.5 Patient Safety

Several strategies may be implemented to ensure patient safety and avoid inappropriate interventions including use of temporary tattoos stating, e.g., “stop save the vein”, identification of vein preservation requirement on iSOFT Clinical Manager handover tab, and identification at the bedside.

Related Documents

- [Government of Western Australia \(2017\) Guideline for Vein Preservation in Chronic Kidney Disease](#)
- [Peripheral Intravascular Catheters \(PIVC\) – Insertion, Maintenance and Removal \(SCGOPHCG\)](#)

References

1. Levey AS, de Jong PE, Coresh J, El Nahas M, Astor BC, Matsushita K, et al. The definition, classification, and prognosis of chronic kidney disease: a KDIGO Controversies Conference report. *Kidney international*. 2011 Jul; 80(1):17-28. PubMed PMID: 21150873.
2. Government of Western Australia. (2017) Guideline for Vein Preservation in Chronic Kidney Disease. Department of Health, Renal Health Network. Perth. Australia

Authorisation & Version Control

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1: Vein Selection Hierarchy

